PHD125 Pharmaceutical Organic Chemistry · Organic Compounds that Contain Oxygen and Sulphur

Before class: one oxygen changes everything

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 15 minutes. Read the three short cards, then try the readiness check. Wrong answers are fine: they show what to listen for in the lecture.

Read the cards first, then answer the questions without notes. The answer key is at the back.

1 · Read first (5 min)

Four families, one atom

Put an **–OH** on a carbon chain and you have an **alcohol** (ethanol). Put an O **between** two carbons: an **ether**. Swap the O of an alcohol for S: a **thiol**. Join C and O with a double bond: a **carbonyl** group, found in **aldehydes** and **ketones**.

Hydrogen bonds again

An –OH group lets molecules hydrogen bond to each other and to water. That is why ethanol (46 g/mol) is a liquid that mixes with water, while propane (44 g/mol) is a gas that does not. Watch for this idea all chapter.

A new kind of reaction

In Chapter 2, reagents **added** to C=C bonds. In this chapter you meet **substitution** by a **nucleophile**: an electron-rich particle such as OH^- swaps places with a halogen. This is how a haloalkane becomes an alcohol, and how some anticancer drugs attach to DNA.

2 · Readiness check

Ten quick questions: recap and a preview.

Q1 Recap

Which intermolecular force is strongest?

A. dispersion
B. dipole–dipole
C. hydrogen bonding
D. they are all equal

Q2 Recap

Propene + H_2O (acid catalyst) gives mainly:

A. 1-propanol
B. 2-propanol
C. propane
D. propanal

Q3 Recap

Which bond is polar?

A. C–H
B. C–C
C. C–O
D. H–H

Q4 CH3-LO1

Match each functional group to its family.

Options: alcohol · ether · thiol · aldehyde · ketone

R–OH _____

R–O–R′ _____

R–SH _____

R–CHO _____

Q5 CH3-LO1

Which ending do alcohol names have? Type it with a hyphen, e.g. -ane.

_____

Q6 CH3-LO1

Ethanol dissolves in water because it can hydrogen bond with water molecules.

True / False

Q7 CH3-LO2

A nucleophile is:

A. an electron-rich species that attacks a δ+ carbon
B. a positive ion that removes electrons
C. a catalyst
D. a noble gas

Q8 CH3-LO3

Which is in hand sanitiser?

A. ethanol
B. ethanal
C. ethane
D. ethanethiol

Q9 CH3-LO5

Acetone (nail-polish remover) belongs to which family?

A. alcohol
B. aldehyde
C. ketone
D. ether

Q10 CH3-LO4

LPG and natural gas have no smell, yet you can smell a leak. Guess what is added and why.

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Adapted from *The Basics of GOB Chemistry* (Ball, Hill & Scott) chapter 14 and exercises 14.E, and *Fundamentals of General Organic and Biological Chemistry* 14.4–14.9 and 15.1–15.5, LibreTexts, CC BY-NC-SA 3.0. SN1 and SN2 items, colour tests and pharmacy scenarios are original material by Dr. NAM. Shared under CC BY-NC-SA for non-commercial teaching.

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About 15 minutes. Read the three short cards, then try the readiness check. Wrong answers are fine: they show what to listen for in the lecture.

Answer key

2 · Readiness check

Q1 Recap

Which intermolecular force is strongest?

A. dispersion

B. dipole–dipole

C. hydrogen bonding

D. they are all equal

Hydrogen bonds (O–H, N–H, F–H) are the strongest intermolecular forces.

Q2 Recap

Propene + H_2O (acid catalyst) gives mainly:

A. 1-propanol

B. 2-propanol

C. propane

D. propanal

Markovnikov's rule (Chapter 2).

Q3 Recap

Which bond is polar?

A. C–H

B. C–C

C. C–O

D. H–H

O is more electronegative than C.

Q4 CH3-LO1

Match each functional group to its family.

R–OH → alcohol; R–O–R′ → ether; R–SH → thiol; R–CHO → aldehyde

Ketone is the extra option: R–CO–R′.

Q5 CH3-LO1

Which ending do alcohol names have? Type it with a hyphen, e.g. -ane.

-ol

Ethanol, propanol. Do not confuse with -al (aldehydes).

Q6 CH3-LO1

Ethanol dissolves in water because it can hydrogen bond with water molecules.

True / False **→ True**

Small alcohols (1–3 C) mix completely with water.

Q7 CH3-LO2

A nucleophile is:

A. an electron-rich species that attacks a δ+ carbon

B. a positive ion that removes electrons

C. a catalyst

D. a noble gas

Examples: OH^-, H_2O, CN^-.

Q8 CH3-LO3

Which is in hand sanitiser?

A. ethanol

B. ethanal

C. ethane

D. ethanethiol

60–80% ethanol or 2-propanol.

Q9 CH3-LO5

Acetone (nail-polish remover) belongs to which family?

A. alcohol

B. aldehyde

C. ketone

D. ether

C=O between two carbons: propanone.

Q10 CH3-LO4

LPG and natural gas have no smell, yet you can smell a leak. Guess what is added and why.

A small amount of a thiol such as ethanethiol, which has a very strong smell, is added so leaks can be detected. Thiols are the sulfur relatives of alcohols (3.2).

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